SPECTROSCOPIC APPLICATIONS OF THE MICHELSON INTERFEROMETER

OPTICS LABORATORY REPORT

Wavelength Measurement and
Fringe-Visibility Analysis

Submitted by

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Code repository: github.com/ParthaXD/Interferrometers

Light Sources:	He–Ne laser and single 533.3 nm source
Instrument:	Michelson interferometer
Measurements:	Laser wavelength and fringe visibility

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Abstract

The Michelson interferometer is a classic amplitude-splitting instrument in which the interference of coherent beams provides a sensitive probe of path difference and of the coherence of a light source. In this work the instrument is used to study two complementary aspects of two-beam interference. In the first, a He–Ne laser is used to produce concentric circular fringes, and the wavelength of the source is determined from the number of fringes that cross the centre of the pattern as the movable mirror is displaced by a known amount. The direct estimate is cross-checked against a least-squares analysis of the cumulative fringe count, and possible systematic contributions such as counting bias, backlash and scale calibration are considered. In the second, a single source with stated wavelength 533.3 nm is used and the contrast of the fringes is examined as a function of mirror displacement. Photographs of the fringe pattern are reduced to luma profiles, from which the visibility is obtained for selected peak–valley pairs, and its dependence on displacement is characterised by a linear model supported by weighted, non-parametric and resampling tests and a full propagation of uncertainties. The relevance of temporal coherence, the source spectrum and the recording conditions to the observed contrast is discussed, together with the limitations of the experimental design and suggestions for improvement.

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1 Introduction

This report examines two complementary Michelson-interferometer measurements: wavelength determination with a He–Ne laser and fringe-visibility analysis using photographs of a single source specified at 533.3 nm.

1.1 Aims and objectives

Wavelength measurement

1. Align the interferometer so that M_1 is perpendicular to M_2 (i.e. $M_1 \parallel M_2'$) and obtain concentric circular fringes on the screen.
2. Record the coarse and fine scale readings of the movable mirror and the number of fringes N that pass the centre for each displacement.
3. Calculate Δd and λ for every step, then the mean λ, and also the wavelength from the total fringe count.
4. Use a least-squares straight-line fit of cumulative fringe count versus mirror displacement as an independent consistency check.
5. Compare the result with the given value of wavelength in the lab manual, $\lambda_0 = 632.8$ nm, estimate the uncertainty, and discuss the sources of error.

Fringe-visibility analysis

1. Obtain a clear interference pattern with the Michelson interferometer and photograph it at 42 different positions of the movable mirror M_1.
2. Extract the luma profile $L(x)$ along a line across the fringes for every photograph.
3. On each profile locate three bright-fringe maxima and the dark-fringe minima that precede them, and compute the visibility of each peak–valley pair.
4. Average the three pair visibilities to obtain one value V_{avg} per photograph and estimate its uncertainty.
5. Plot V_{avg} against z and fit a straight line; assess the quality of the fit and the statistical significance of the slope.
6. Discuss, separating what the data support from what is only a possible explanation, the reasons for the observed behaviour.

2 Instrument and optical arrangement

Common apparatus

- Michelson interferometer with fixed mirror M_2, movable mirror M_1 on a translation stage, beam splitter (BS), compensating plate (CP) and tilt screws for both mirrors.
- Coarse adjustment knob (least count 0.01 mm) and fine adjustment knob (least count 0.0001 mm).
- White viewing screen, placed about 1–2 m from M_1 for better resolution.

Wavelength measurement

- He–Ne laser ($\lambda_0 = 632.8$ nm, the given value of wavelength in the lab manual) mounted on a lab jack.
- Beam expander (microscope objective) with holder.

Fringe-visibility analysis

- Single source for the fringe-visibility measurement, with stated wavelength $\lambda = 533.3$ nm, and a ground-glass screen in front of it to reduce its brightness.
- Mobile Camera used to photograph the fringes.
- Computer with Python (`numpy`, `scipy`, `matplotlib`) for the analysis.

3 Physical basis and measurement models

3.1 Working principle

The Michelson interferometer is an amplitude-splitting interferometer. Light from the source S falls on a beam splitter, whose rear face is semi-silvered, so that roughly half of the light is reflected towards the movable mirror M_1 and the other half is transmitted towards the fixed mirror M_2 . Both beams are reflected back to the beam splitter and leave towards the observation point E, where they overlap. Because both beams come from the same source they are coherent, so a stable pattern of bright and dark fringes is seen. Moving M_1 changes the optical path of one arm and therefore shifts the pattern.

3.2 Compensating plate

The beam travelling to M_1 passes through the glass of the beam splitter twice, while the beam going to M_2 would travel entirely in air if the compensating plate were absent. This introduces an extra optical path

$$\delta = 2(\mu - 1)t, \tag{1}$$

where t is the plate thickness and μ its refractive index. For a single-wavelength source such as a laser this only shifts the pattern, so the compensating plate is not essential. For white light it matters, since μ depends on wavelength, and an identical plate CP is placed in the second arm.

3.3 Circular fringes of equal inclination and the path difference

Let d be the separation between M_1 and the virtual image M_2'. When $M_1 \parallel M_2'$ the two coherent virtual images of the source are separated by $2d$, and for rays leaving at angle θ to the axis the path difference is

$$\Delta = 2d\cos\theta. \tag{2}$$

A bright fringe is formed when

$$2d\cos\theta = n\lambda \qquad (n = 0, 1, 2, \dots), \tag{3}$$

and a dark fringe when $2d\cos\theta = (n+\frac{1}{2})\lambda$. For fixed n, λ and d the angle θ is fixed, so every fringe is a circle about the line through the observer perpendicular to the mirrors, which is why these are called fringes of equal inclination. If M_1 and M_2' are slightly inclined the air film is a wedge and the fringes are curved (equal thickness); if they intersect the fringes are straight [1]. Many photographs from the fringe-visibility analysis show elliptical or curved fringes, which indicates a slightly wedge-shaped film or an oblique view.

Moving M_1 by a distance z changes d by z and therefore changes the optical path difference on axis by $2z$. For the fringe-visibility analysis, z is the displacement of M_1 from its position at the first photograph, read from the coarse knob. The absolute value of d at $z = 0$ was not recorded, so only changes in path difference ($2z$) are known.

3.4 Wavelength from fringe counting

At the centre of the pattern $\theta = 0$, so Eq. (3) reduces to $2d = n\lambda$. If the mirror is moved by Δd and N fringes appear (or disappear) at the centre, the order changes from n to $n + N$:

$$2(d + \Delta d) = (n + N)\lambda \implies \lambda = \frac{2\,\Delta d}{N}. \tag{4}$$

Every fringe crossing the centre therefore corresponds to a mirror motion of only $\lambda/2$.

3.5 Expected fringe count (sanity check)

For $\lambda_0 = 632.8$ nm one fringe corresponds to $\lambda_0/2 = 316.4$ nm of mirror motion. A displacement of one main-scale division, 0.010 mm = 10 000 nm, should therefore produce

$$N_{\text{exp}} = \frac{2\,\Delta d}{\lambda_0} = \frac{2 \times 0.010\ \text{mm}}{632.8 \times 10^{-6}\ \text{mm}} \approx 31.6 \tag{5}$$

fringes, and a total travel of 0.100 mm should produce about 316.1 fringes. We use this number to judge whether our counts are of the right size.

3.6 Position of the mirror from the two knobs

With p the coarse (main-scale) reading and q the fine (circular-scale) reading, the position of M_1 is

$$d = p \times 0.01 \text{ mm} + q \times 0.0001 \text{ mm}. \tag{6}$$

3.7 Least-squares fitting

Wavelength measurement (cumulative fringe count). If M_k is the cumulative number of fringes counted after the mirror has moved a total distance x_k from the starting position, then Eq. (4) implies a straight line

$$M(x) = \frac{2}{\lambda}\, x + M_0, \tag{7}$$

so that the slope is $s = 2/\lambda$ and $\lambda = 2/s$. The intercept M_0 absorbs any fringes lost or gained at the very start (for instance, through backlash in the screw). The relative uncertainty of the wavelength obtained from the slope is $\delta\lambda/\lambda = \delta s/s$.

General formulas (used in both measurements). For n points (z_i, V_i) fitted by $V = a+bz$, with $\bar{z}$ and $\bar{V}$ the means, $S_{zz} = \sum(z_i - \bar{z})^2$, $S_{zV} = \sum(z_i - \bar{z})(V_i - \bar{V})$ and $S_{VV} = \sum(V_i - \bar{V})^2$:

$$b = \frac{S_{zV}}{S_{zz}}, \qquad a = \bar{V} - b\bar{z}, \tag{8}$$

$$s^2 = \frac{\sum_i (V_i - a - bz_i)^2}{n-2}, \qquad \delta b = \frac{s}{\sqrt{S_{zz}}}, \qquad \delta a = s\sqrt{\frac{1}{n} + \frac{\bar{z}^2}{S_{zz}}}, \tag{9}$$

$$r = \frac{S_{zV}}{\sqrt{S_{zz}S_{VV}}}, \qquad R^2 = r^2, \qquad R^2_{\text{adj}} = 1 - (1-R^2)\frac{n-1}{n-2}, \tag{10}$$

$$t = \frac{b}{\delta b}, \qquad b \pm t_{0.975,\,n-2}\,\delta b \;\; (95\%\text{ CI}), \qquad F = t^2. \tag{11}$$

The p-value is the probability of a $|t|$ at least as large as observed if the true slope were zero. At position z, the 95% confidence band of the fitted line has half-width

$$t_{0.975,n-2}\, s\sqrt{1/n + (z-\bar{z})^2/S_{zz}}.$$

If individual uncertainties σ_i are available, the weighted fit minimises $\chi^2 = \sum_i (V_i - a - bz_i)^2/\sigma_i^2$ with weights $w_i = 1/\sigma_i^2$, and the reduced chi-square $\chi^2_\nu = \chi^2/(n-2)$ shows whether the scatter around the line is consistent with the stated uncertainties ($\chi^2_\nu \approx 1$) or much larger ($\chi^2_\nu \gg 1$).

Residuals of a fit are only independent if the model describes the data. The Durbin–Watson statistic $DW = \sum(e_i - e_{i-1})^2 / \sum e_i^2$ is about 2 for uncorrelated residuals and well below 2 for positively correlated residuals. If the lag-1 autocorrelation of the residuals is ρ, the effective number of independent points is approximately $n_{\text{eff}} = n(1-\rho)/(1+\rho)$ and the slope uncertainty should be multiplied by about $\sqrt{n/n_{\text{eff}}}$.

3.8 Visibility of two-beam fringes

In a Michelson interferometer a beam splitter divides the light into two beams which travel to M_1 and M_2 and recombine. If the two beams have intensities I_1 and I_2 and a phase difference δ, the resulting intensity is

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\,|\gamma|\cos\delta, \tag{12}$$

where $|\gamma|$ ($0 \le |\gamma| \le 1$) is the degree of coherence of the two beams. The maximum and minimum intensities are $I_{\max} = I_1 + I_2 + 2\sqrt{I_1 I_2}|\gamma|$ and $I_{\min} = I_1 + I_2 - 2\sqrt{I_1 I_2}|\gamma|$. The *visibility* (contrast) of the fringes is

$$V = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} = \frac{2\sqrt{I_1 I_2}}{I_1 + I_2}\,|\gamma|. \tag{13}$$

Thus $V = 1$ for perfect fringes ($I_{\min} = 0$) and $V = 0$ when the fringes have disappeared. Equation (13) shows three ways in which V can fall: (i) the two beams become unequal in intensity, (ii)

the coherence $|\gamma|$ falls, (iii) the recorded pattern contains extra background light, or is smeared, so that $I_{\min}$ rises relative to $I_{\max}$.

3.9 Interference with a finite coherence time

Let the two beams reaching the observation point be $E_1 = E_0 e^{i(kl_1-\omega t)}$ and $E_2 = E_0 e^{i(kl_2-\omega t)}$, where l_1, l_2 are the optical path lengths of the two arms. With $I_0 = \frac{1}{2}\epsilon_0 c|E_0|^2$ the intensity of one beam,

$$I = \tfrac{1}{2}\epsilon_0 c\,|E_1 + E_2|^2 = 2I_0\,[1 + \cos k(l_1 - l_2)]\,. \tag{14}$$

A real source is not a perfect monochromatic wave: its phase wanders, so the field at a point is $E(t)$ with a finite coherence time. The beam that has travelled the longer path arrives a time $\tau = (l_1 - l_2)/c = \Delta/c$ later, and the intensity is

$$I = \tfrac{1}{2}\epsilon_0 c\,\big\langle\, |E(t) + E(t+\tau)|^2 \big\rangle = 2I_0\,[1 + \operatorname{Re}\gamma(\tau)]\,, \tag{15}$$

where $\langle \dots \rangle$ is a time average, $\Gamma(\tau) = \langle E(t)E^*(t+\tau)\rangle$ is the autocorrelation function of the field, and

$$\gamma(\tau) = \frac{\Gamma(\tau)}{\Gamma(0)}, \qquad 0 \le |\gamma| \le 1 \tag{16}$$

is the complex degree of (temporal) coherence. Writing $\gamma = |\gamma| e^{i\phi(\tau)}$, the intensity oscillates as $2I_0[1 + |\gamma|\cos\phi]$ between

$$I_{\max} = 2I_0\,[1 + |\gamma(\tau)|]\,, \qquad I_{\min} = 2I_0\,[1 - |\gamma(\tau)|]\,, \tag{17}$$

so that the visibility is

$$V = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} = |\gamma(\tau)|. \tag{18}$$

The visibility of the fringes is therefore a direct measure of the degree of temporal coherence at the delay $\tau = 2z/c$ set by the mirror displacement. The delay beyond which $|\gamma|$ has fallen appreciably is the coherence time τ_c, and $\ell_c = c\,\tau_c$ is the coherence length; fringes are visible only while the path difference is not much larger than ℓ_c. By the Wiener–Khinchin theorem $\Gamma(\tau)$ is the Fourier transform of the power spectrum of the source, so a narrower spectrum means a longer coherence length.

3.10 Coherence length and spectral width

For a single source with a narrow spectral width $\Delta\lambda$ about central wavelength λ, the coherence length is of order

$$\ell_c \sim \frac{\lambda^2}{\Delta\lambda}. \tag{19}$$

The central wavelength alone does not determine ℓ_c; the source linewidth or a well-resolved visibility envelope is also required. Here $\lambda = 533.3$ nm, but no linewidth is supplied. The visibility data can therefore provide only an operational range over which fringes remain measurable, not a source linewidth or a precise intrinsic coherence length. Section 6.4.3 discusses the conditional estimate from the observed scan.

3.11 Luma and intensity

The photographs are 8-bit images. The *luma* L (0–255) is a weighted sum of the colour channels (for example $L = 0.299R + 0.587G + 0.114B$). Camera images are gamma-encoded, so luma is a non-linear function of the light intensity, approximately

$$L \approx 255\left(\frac{I}{I_{\text{sat}}}\right)^{1/\gamma_c}, \qquad \gamma_c \approx 2.2. \tag{20}$$

The visibility computed directly from luma values,

$$V_L = \frac{L_{\max} - L_{\min}}{L_{\max} + L_{\min}}, \tag{21}$$

is therefore not identical to the intensity visibility of Eq. (13); it is smaller, because the gamma curve compresses the range between bright and dark. The quantity V_L is used throughout this report as the measure of contrast (it is monotonic in the true contrast), and Section 6.3.11 checks how the result changes if the luma is converted back to linear intensity using Eq. (20).

3.12 Visibility from a peak–valley pair

On a profile $L(x)$ across the fringes, each peak is one bright fringe and each valley is one dark fringe. For a peak of height $L_{\max}$ and the valley of height $L_{\min}$ chosen with it,

$$V_k = \frac{L_{\max}^{(k)} - L_{\min}^{(k)}}{L_{\max}^{(k)} + L_{\min}^{(k)}}, \qquad k = 1, 2, 3, \tag{22}$$

and the visibility of the photograph is the mean of the three values,

$$V_{\text{avg}} = \frac{V_1 + V_2 + V_3}{3}. \tag{23}$$

Pairing each peak with a valley *adjacent* to it, instead of comparing the highest peak of the whole profile with its lowest valley, avoids counting the slow variation of the beam brightness across the screen as fringe contrast. In this report the valley chosen for each peak is the lowest point lying *before* that peak (between the previous peak and this one; for the first peak, from the start of the profile).

3.13 Propagation of uncertainty in V

Write $V = (a - b)/(a + b)$ with $a = L_{\max}$ and $b = L_{\min}$. Then

$$\frac{\partial V}{\partial a} = \frac{2b}{(a+b)^2}, \qquad \frac{\partial V}{\partial b} = -\frac{2a}{(a+b)^2}, \tag{24}$$

so for independent uncertainties $\delta a, \delta b$,

$$\delta V = \frac{2}{(a+b)^2}\sqrt{b^2\,\delta a^2 + a^2\,\delta b^2}. \tag{25}$$

For the mean of three pair values, if the pair uncertainties are independent,

$$\delta V_{\text{avg}}^{\text{prop}} = \frac{1}{3}\sqrt{\delta V_1^2 + \delta V_2^2 + \delta V_3^2}. \tag{26}$$

4 Experimental method

4.1 Alignment and observation of circular fringes (both measurements)

1. The interferometer was placed on the table with the coarse knob facing the observer.
2. The lab jack was set in front of the objective holder and its height adjusted.
3. The He–Ne laser was placed on the lab jack and pointed at the centre of the fixed mirror M_2.
4. The laser was switched on and its height adjusted until the beam ran roughly parallel to the top plate of the interferometer and hit the mirror at its centre.
5. The screen was placed opposite M_1, about 1–2 m away, for better resolution.
6. Using the coarse knob, the distances of M_1 and M_2 from the beam splitter were made nearly equal.

7. Several spots appeared on the screen: two faint ones (reflection from the un-silvered face of the BS, then from M_1 and M_2) and two bright ones (from the silvered face). Two extra spots from the compensating plate were ignored.
8. The tilt screws at the back of M_1 and M_2 were adjusted until only two spots remained, i.e. until $M_1 \perp M_2$ and $M_1 \parallel M_2'$.
9. The beam expander was inserted in front of the laser, its height adjusted, and the top tilt screws were used for fine adjustment until clear, concentric circular fringes were visible on the screen.

4.2 Wavelength measurement

1. With the circular fringes steady, the mirror was moved using the knob and the fringes were seen to appear (or vanish) at the centre. The knob was turned in one direction only throughout, to avoid backlash.
2. The coarse reading p and fine reading q were noted, and the starting position d_1 was obtained from Eq. (6). In our run the starting reading was $p = 17$, $q = 30$.
3. The knob was then turned so that the mirror moved through exactly one main-scale division, and the number of fringes N crossing the centre was counted.
4. The new readings (p, q) were noted to obtain d_2, and $\Delta d = d_2 - d_1$ was calculated.
5. This was repeated for ten successive divisions ($p = 18$ to 27), keeping the fine reading fixed at $q = 30$. The end position of one step served as the start of the next.
6. λ was calculated from Eq. (4) for each step and for the whole run.

4.3 Recording the single-source fringes

1. The He–Ne laser was replaced by the single Part 2 source ($\lambda = 533.3$ nm), with a ground-glass screen in front of it to reduce its brightness. The interferometer was kept aligned so that the mirrors M_1 and M_2' were nearly parallel and a clear fringe pattern was visible on the screen (alignment as in Section 4.1).
2. The initial reading of the movable mirror was taken as $z = 0$ and the first photograph was recorded.
3. The coarse knob was then turned, always in the same direction to avoid backlash, to increase the mirror displacement, and a photograph was taken at each new position.
4. The displacement was increased in steps of 0.2 mm from $z = 0$ to 6.4 mm (photographs 1 to 33) and in steps of 0.4 mm from 6.4 to 10.0 mm (photographs 34 to 42). In total 42 photographs were recorded.

The camera settings, screen distance and mirror tilt were not logged separately for every photograph, so small changes in framing or alignment cannot be ruled out.

4.4 Extracting the luma profiles

For each photograph, a line profile was extracted across several visible fringes. The pixel coordinate along that line served as the profile coordinate, and the corresponding luma value formed the measured trace. Three bright maxima and the dark minima preceding them were identified after light smoothing; their coordinates and values were retained for every photograph in the complete peak–valley table (Table 8).

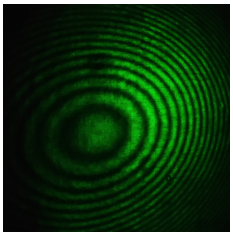


Figure 1: Photograph 42: example fringe pattern from the Part 2 source.

The raw and lightly smoothed traces were compared during peak selection so that the selected extrema remained tied to the observed profile. The three pair visibilities were then calculated separately and averaged to produce one value per image. Photograph 42 is included as the single visual example of the fringe pattern; the remaining photographs contribute through their recorded profile data and are not reproduced as separate illustrations.

Camera settings, screen distance and mirror tilt were not logged separately for every photograph. Small differences in framing or alignment therefore remain possible sources of between-image variation. The luma-based result is a relative contrast index, not a calibrated photometric visibility.

5 Laser wavelength measurement

5.1 Least counts

Coarse scale: 0.01 mm per division. Fine scale: 0.0001 mm per division. Reference wavelengths used in this report: $\lambda_{\text{He–Ne}} = 632.8$ nm for Part 1 and $\lambda_{\text{Part 2}} = 533.3$ nm for the single source used in the visibility measurement.

5.2 Recorded data

Table 1 reproduces the readings exactly as written in the lab notebook (data set 3, signed on 2 September 2026). Row 1 is the starting position, so no fringes were counted for it. For each later row, N is the number of fringes counted while the mirror moved from the previous position to the position in that row; hence d_1 of a row equals d_2 of the row above.

Table 1: Raw readings of the coarse scale p, fine scale q and fringe count N. Positions follow Eq. (6).

Sl. no.	N	d_1 (mm)	p	q	$d_2 = 0.01p + 0.0001q$ (mm)
1	–	–	17	30	0.173
2	30	0.173	18	30	0.183
3	31	0.183	19	30	0.193
4	31	0.193	20	30	0.203
5	31	0.203	21	30	0.213
6	31	0.213	22	30	0.223
7	31	0.223	23	30	0.233
8	31	0.233	24	30	0.243
9	31	0.243	25	30	0.253
10	32	0.253	26	30	0.263
11	31	0.263	27	30	0.273
Total: $\sum N = 310$ fringes; $d_{\text{final}} - d_{\text{initial}} = 0.273 - 0.173 = 0.100$ mm					

Check of the position formula. For the starting position, $d = 17 \times 0.01 + 30 \times 0.0001 = 0.170 + 0.003 = 0.173$ mm. For the last row, $d = 27 \times 0.01 + 30 \times 0.0001 = 0.270 + 0.003 = 0.273$ mm, as recorded. Because q was the same in every row, each step is exactly $\Delta d = 0.010$ mm.

5.3 Wavelength calculation and data analysis

5.3.1 Displacement and wavelength for each step

For each of the ten steps, $\Delta d = d_2 - d_1$ and, from Eq. (4),

$$\lambda_i = \frac{2\,\Delta d}{N_i} = \frac{2 \times 0.010 \text{ mm}}{N_i} = \frac{20\,000 \text{ nm}}{N_i}.$$

Worked examples.

- Sl. no. 2 ($N = 30$): $\Delta d = 0.183 - 0.173 = 0.010$ mm, so $\lambda = \dfrac{2 \times 0.010}{30}$ mm $= 6.667 \times 10^{-4}$ mm $= 666.7$ nm.
- Sl. no. 3 ($N = 31$): $\Delta d = 0.193 - 0.183 = 0.010$ mm, so $\lambda = \dfrac{2 \times 0.010}{31}$ mm $= 6.452 \times 10^{-4}$ mm $= 645.2$ nm.

- Sl. no. 10 ($N = 32$): $\Delta d = 0.263 - 0.253 = 0.010$ mm, so $\lambda = \dfrac{2 \times 0.010}{32}$ mm $= 6.250 \times 10^{-4}$ mm $= 625.0$ nm.

The other rows follow in the same way and are collected in Table 2. The last column gives the deviation of each single-step value from the given value of wavelength in the lab manual.

Table 2: Step-by-step wavelength from Eq. (4) ($\lambda_0 = 632.8$ nm).

Sl. no.	N	d_1 (mm)	d_2 (mm)	Δd (mm)	λ_i (nm)	$\lambda_i - \lambda_0$ (nm)
2	30	0.173	0.183	0.010	666.7	+33.9
3	31	0.183	0.193	0.010	645.2	+12.4
4	31	0.193	0.203	0.010	645.2	+12.4
5	31	0.203	0.213	0.010	645.2	+12.4
6	31	0.213	0.223	0.010	645.2	+12.4
7	31	0.223	0.233	0.010	645.2	+12.4
8	31	0.233	0.243	0.010	645.2	+12.4
9	31	0.243	0.253	0.010	645.2	+12.4
10	32	0.253	0.263	0.010	625.0	−7.8
11	31	0.263	0.273	0.010	645.2	+12.4
Total				0.100	–	–

5.3.2 Mean of the individual wavelengths

The manual asks for the mean of the individual λ values:

$$\bar{\lambda} = \frac{1}{10}\sum_i \lambda_i = \frac{666.67 + 8(645.16) + 625.00}{10} = \frac{6452.96}{10} = 645.30 \text{ nm}.$$

The sample standard deviation is

$$\sigma_\lambda = \sqrt{\frac{\sum(\lambda_i - \bar{\lambda})^2}{n-1}} = \sqrt{\frac{(21.37)^2 + (-20.30)^2 + 8(-0.135)^2}{9}} = \sqrt{\frac{868.8}{9}} = 9.83 \text{ nm},$$

and the standard error of the mean is $\sigma_\lambda/\sqrt{10} = 9.83/3.162 = 3.11$ nm. So $\bar{\lambda} = 645.3 \pm 3.1$ nm.

5.3.3 Wavelength from the total fringe count

A single step gives only a coarse estimate: a miscount by one fringe in about 31 changes λ by about 3%. It is better to add all the counts first and divide once, because a ± 1 error then matters much less. The total count is

$$N_{\text{tot}} = 30+31+31+31+31+31+31+31+32+31 = 310, \qquad \Delta d_{\text{tot}} = 0.273 - 0.173 = 0.100 \text{ mm}.$$

Therefore

$$\lambda = \frac{2\,\Delta d_{\text{tot}}}{N_{\text{tot}}} = \frac{2 \times 0.100 \text{ mm}}{310} = 6.452 \times 10^{-4} \text{ mm} = 645.2 \text{ nm}.$$

Each fringe thus corresponds to a mirror motion of 0.100 mm/310 = 322.6 nm, compared with the expected 316.4 nm.

Statistical uncertainty. The per-step counts have mean $\bar{N} = 310/10 = 31.00$ and sample standard deviation

$$\sigma_N = \sqrt{\frac{(30-31)^2 + 8(31-31)^2 + (32-31)^2}{10-1}} = \sqrt{\frac{2}{9}} = 0.471,$$

so the standard error is $\sigma_N/\sqrt{10} = 0.149$. Since $\lambda \propto 1/\bar{N}$,

$$\frac{\delta\lambda}{\lambda} = \frac{0.149}{31.00} = 0.481\% \;\Rightarrow\; \delta\lambda = 645.2 \times 0.00481 = 3.1 \text{ nm}, \qquad \lambda = 645.2 \pm 3.1 \text{ nm (statistical)}.$$

5.3.4 Comparison with the expected count

From Eq. (5) we expect 31.61 fringes per division, i.e. 316.1 fringes for 0.100 mm, against 310 counted. The shortfall is

$$316.1 - 310 = 6.1 \text{ fringes } (\approx 1.9\%),$$

or 0.61 fringe per division on average. The observed counts (30–32) are thus of the right size, but all ten of them are at or slightly below the expected 31.6 (in terms of the nearest whole number, none exceeds 32). That already hints that the deviation is systematic rather than random.

5.3.5 Consistency check: linear fit of the cumulative count

To avoid relying on a single ratio, the cumulative count $M_k = \sum_{i \le k} N_i$ was plotted against the cumulative displacement $x_k = 0.01\,k$ mm and fitted with Eq. (7). The fit was done in terms of the step index k (so $M_k = s_k\,k + c$) and the slope was converted to a per-mm value afterwards. Table 3 lists the quantities needed. Here $k = \text{Sl. no.} - 1$.

Table 3: Quantities for the least-squares fit, with fitted values $M_k^{\text{fit}} = 31.097\,k - 1.333$ and residuals $r_k = M_k - M_k^{\text{fit}}$.

k	x_k (mm)	N_k	M_k	k^2	kM_k	M_k^{fit}	r_k	r_k^2
1	0.01	30	30	1	30	29.764	+0.236	0.0559
2	0.02	31	61	4	122	60.861	+0.139	0.0194
3	0.03	31	92	9	276	91.958	+0.042	0.0018
4	0.04	31	123	16	492	123.055	−0.055	0.0030
5	0.05	31	154	25	770	154.152	−0.152	0.0229
6	0.06	31	185	36	1110	185.248	−0.248	0.0618
7	0.07	31	216	49	1512	216.345	−0.345	0.1194
8	0.08	31	247	64	1976	247.442	−0.442	0.1957
9	0.09	32	279	81	2511	278.539	+0.461	0.2122
10	0.10	31	310	100	3100	309.636	+0.364	0.1322
Σ	–	310	1697	385	11899	–	0.000	0.8242

With $n = 10$, $\sum k = 55$, $\sum k^2 = 385$, $\sum M = 1697$ and $\sum kM = 11899$:

$$S_{kk} = n\sum k^2 - \Big(\sum k\Big)^2 = 10(385) - 55^2 = 825,$$
$$S_{kM} = n\sum kM - \sum k \sum M = 10(11899) - 55(1697) = 25655.$$

Slope and intercept:

$$s_k = \frac{S_{kM}}{S_{kk}} = \frac{25655}{825} = 31.097 \text{ fringes per step},$$
$$c = \frac{\sum M - s_k \sum k}{n} = \frac{1697 - 31.097(55)}{10} = -1.33 \text{ fringes}.$$

Dividing the slope by the step size (0.01 mm):

$$s = \frac{31.097}{0.01 \text{ mm}} = 3109.7 \text{ mm}^{-1}, \qquad \lambda_{\text{fit}} = \frac{2}{s} = \frac{2}{3109.7 \text{ mm}^{-1}} = 6.431 \times 10^{-4} \text{ mm} = 643.1 \text{ nm}.$$

(More precisely 643.15 nm, which we quote as 643.2 nm below.)

Uncertainties of the fit. The residual sum of squares is $\sum r_k^2 = 0.824$ (Table 3), so the residual standard deviation is

$$s_r = \sqrt{\frac{\sum r_k^2}{n-2}} = \sqrt{\frac{0.824}{8}} = 0.321 \text{ fringe}.$$

With $\sum(k-\bar{k})^2 = S_{kk}/n = 82.5$ and $\bar{k} = 5.5$:

$$\delta s_k = \frac{s_r}{\sqrt{82.5}} = \frac{0.321}{9.083} = 0.035, \qquad \delta c = s_r\sqrt{\frac{1}{n} + \frac{\bar{k}^2}{82.5}} = 0.321\sqrt{0.1 + 0.3667} = 0.22.$$

Hence

$$M(x) = (3109.7 \pm 3.5)\,x - (1.33 \pm 0.22), \quad x \text{ in mm}, \quad R^2 = 0.99999,$$

$$\frac{\delta\lambda}{\lambda} = \frac{\delta s}{s} = \frac{3.5}{3109.7} = 0.11\% \;\Rightarrow\; \lambda_{\text{fit}} = 643.2 \pm 0.7 \text{ nm}.$$

The slope is $31.097/31.606 = 0.984$ times the expected value, i.e. 1.6% smaller.

Interpretation of the intercept. A negative intercept of -1.33 ± 0.22 fringes corresponds to 1.33×316.4 nm ≈ 0.42 μm of mirror motion that did not produce a counted fringe at the start. This is of the order that backlash in the drive could produce. It accounts for only about 1.3 of the 6.1 missing fringes, though. The rest (≈ 5 fringes) builds up gradually over the ten steps and shows up as the slightly smaller slope.

A caution on the fit error. The residuals of a cumulative count are not independent, since a miscount in one step shifts all later points. The least-squares formula assumes independent residuals, so the quoted ± 0.7 nm is optimistic and should be treated as a lower limit.

5.3.6 Effect of the first reading

The first step gave $N = 30$, which is one fringe below the steady value of 31. Because the start-up of the drive is where backlash is most likely to act, we checked what happens if this step is left out. The remaining nine steps give $N_{\text{tot}} = 280$ over 0.090 mm:

$$\lambda = \frac{2 \times 0.090 \text{ mm}}{280} = 6.429 \times 10^{-4} \text{ mm} = 642.9 \text{ nm}.$$

Their mean count is 31.11 with standard error 0.111, i.e. 0.36%, so $\delta\lambda = 2.3$ nm and $\lambda = 642.9 \pm 2.3$ nm. The change from 645.2 nm is small and within the uncertainty, so the first step is not responsible for the offset from the given value of wavelength in the lab manual. We kept all ten steps in the main result.

5.3.7 Error analysis

Uncertainty budget for the total-count result. The scatter term and the ± 1 fringe term

Table 4: Contributions to the uncertainty of $\lambda = 645.2$ nm.

Source	Estimate	$\delta\lambda$ (nm)
Scatter of per-step counts (statistical)	$0.149/31.00 = 0.48\%$	3.10
Counting resolution on the total, ± 1 fringe in 310	$1/310 = 0.32\%$	2.08
Reading of d: ± 0.0001 mm on each of two readings	$\sqrt{2}(0.0001)/0.100 = 0.14\%$	0.91
Combined (quadrature)		3.85

partly overlap (the scatter itself comes from counting differences), so adding them in quadrature is a conservative choice. We therefore quote ± 3.1 nm as the statistical uncertainty and ± 3.9 nm as the more cautious combined value.

Deviation from the value given in the lab manual.

$$\lambda - \lambda_0 = 645.2 - 632.8 = +12.4 \text{ nm}, \qquad \text{percentage difference} = \frac{12.4}{632.8} \times 100 = 1.95\%,$$

$$\text{in units of the uncertainty} = \frac{12.4}{3.1} \approx 4.0\,\sigma_{\text{stat}} \quad \text{or} \quad \frac{12.4}{3.85} \approx 3.2\,\sigma_{\text{comb}}.$$

For the fit, $643.2 - 632.8 = +10.4$ nm (1.64%). The given value of wavelength in the lab manual is also outside the 95% confidence interval of the total-count result ($645.2 \pm 2.26 \times 3.1 = 645.2 \pm 7.0$ nm, i.e. 638.2 to 652.2 nm). Because both methods are above 632.8 nm by a similar amount, the discrepancy is statistically significant, and we conclude that it is more likely to be a small systematic effect than random scatter. These are examined in Section 5.6.

Table 5: All estimates of λ compared with the given value 632.8 nm.

Method	λ (nm)	$\lambda - \lambda_0$ (nm)	Difference (%)
Total count, 10 steps (main result)	645.2 ± 3.1	$+12.4$	1.95
Mean of the ten λ_i	645.3 ± 3.1	$+12.5$	1.97
Linear fit of cumulative count	643.2 ± 0.7	$+10.4$	1.64
Total count without step 1 (9 steps)	642.9 ± 2.3	$+10.1$	1.59

5.4 Diagnostic plots for the laser measurement

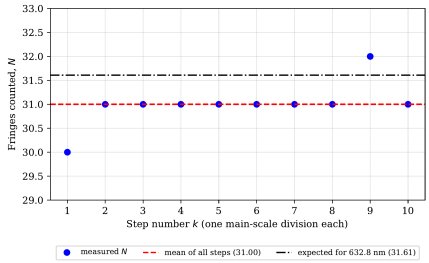


Figure 2: Fringes counted per 0.010 mm step; all ten counts lie within ±1 of the mean.

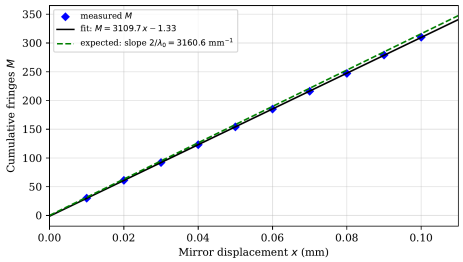


Figure 3: Cumulative fringe count against mirror displacement; the fitted slope gives $\lambda_{\text{fit}} = 643.2$ nm.

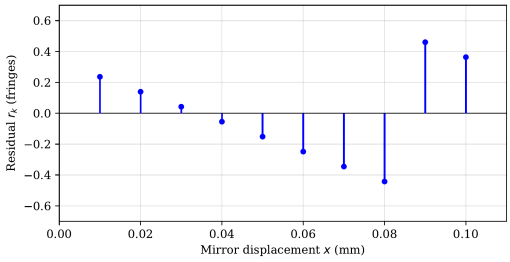


Figure 4: Residuals of the cumulative-count fit, all within ±0.5 fringe.

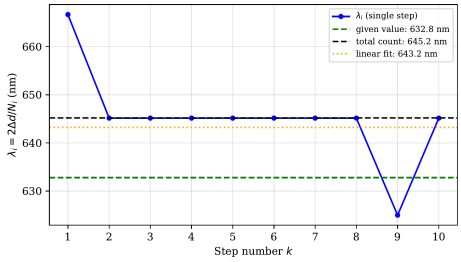


Figure 5: Stepwise wavelength estimates; integer fringe counts permit only three values for each step.

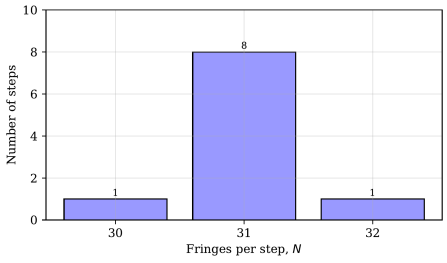


Figure 6: Histogram of the ten fringe counts; eight steps gave 31 fringes.

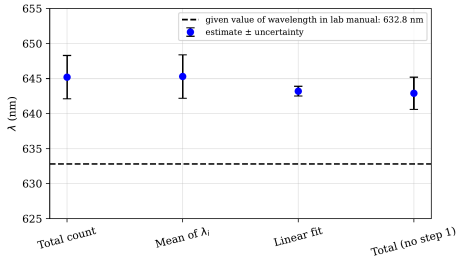


Figure 7: Comparison of the four wavelength estimates and their uncertainties.

5.5 Laser measurement results

Final wavelength result: $\lambda_{\text{He–Ne}} = 645.2 \pm 3.1$ nm (stat.) [±3.9 nm combined]

which is 1.95% above the given value of wavelength in the lab manual, 632.8 nm.

Table 6: Summary of wavelength results.

Quantity	Experimental	Literature / expected	Difference (%)
Total fringes for 0.100 mm	310	316.1	−1.9
Fringes per 0.010 mm (mean)	31.00 ± 0.15	31.61	−1.9
Fringes per 0.010 mm (fit slope)	31.10 ± 0.04	31.61	−1.6
Mirror motion per fringe	322.6 nm	316.4 nm	+2.0
λ (total count)	645.2 ± 3.1 nm	632.8 nm	+1.95
λ (mean of λ_i)	645.3 ± 3.1 nm	632.8 nm	+1.97
λ (linear fit)	643.2 ± 0.7 nm	632.8 nm	+1.64
Fit intercept M_0	-1.33 ± 0.22 fringes	0	–
R^2 of the fit	0.99999	1	–

5.6 Interpretation of the laser result

5.6.1 What the data show

Roughly 31 fringes crossed the centre for every 0.010 mm of mirror movement, in close agreement with the 31.6 expected for a He–Ne laser. The cumulative count is almost perfectly linear ($R^2 = 0.99999$), and the residuals are all below half a fringe. This tells us that the interferometer was well aligned, that the fringes were clear enough to be counted reliably, and that the mirror moved steadily in a single direction. The experiment therefore confirms the relation $\lambda = 2\Delta d/N$ and gives a wavelength that is within about 2% of the accepted value.

5.6.2 Why is the result slightly high?

All four ways of analysing the data give a value between 642.9 and 645.3 nm, i.e. about 10–12 nm above 632.8 nm. A random counting error would scatter some estimates above and some below, so a consistent offset suggests a systematic cause. In terms of fringes, we counted about 6 fewer than expected over 0.100 mm. Plausible contributions are:

- **Under-counting.** At about 31 fringes per division, a fringe at the centre can easily be missed when the pattern moves quickly, especially when brightness varies. Missing even 0.6 fringe per step on average would explain the whole offset. Our counts never exceeded 32, whereas an over-count would have pushed some steps above 31.6.
- **Backlash.** The fit gives an intercept of -1.33 ± 0.22 fringes, meaning that some initial turning of the knob moved the mirror without a counted fringe. This is consistent with backlash, but it explains only about one fifth of the total shortfall.
- **Scale calibration.** If one main-scale division of the coarse knob corresponds to slightly less than 0.010 mm (by about 1.9%), the same shift would result. We had no independent length standard to test this.

Our data cannot tell which of these dominates, or whether several act together, so we do not claim a definite cause.

5.6.3 Comparing the methods

The total-count and mean-λ_i values are practically identical (645.2 and 645.3 nm) because all steps cover the same Δd. The linear fit gives a slightly lower value (643.2 nm) because it allows for a constant offset (intercept) at the start, so it is less affected by backlash than the direct ratio. The fit uncertainty (± 0.7 nm) is much smaller than the others, but because the errors of a cumulative count are correlated, it is too optimistic. The residuals also show a smooth pattern (positive, then negative, then positive) rather than random signs, though with an amplitude below half a fringe. Hence the total-count uncertainty is the more honest figure, and the fit is best used as a check.

5.6.4 Limits of this data set

The run consists of a single set of ten steps (data set 3), all in one direction and covering only 0.100 mm of travel. With so few steps, a systematic effect cannot be separated from a statistical

one, and the sample standard deviation of N rests on only ten values, two of which differ from the rest. A longer run, repeated sets, and an independent count by a second observer would give a much firmer statement about the origin of the 1.9% offset.

6 Single-source fringe-visibility analysis and results

The same automated procedure was applied to every profile. The supplied project did not include its Python source file, so the procedure is documented below rather than reproduced as code.

1. **Smoothing for detection.** The raw profile $y(x)$ (with N_p points) is smoothed with a Gaussian filter of standard deviation $\sigma_d = 0.025N_p$ pixels to suppress speckle and pixel noise.

2. **Peaks.** All local maxima of the smoothed curve are found and ranked by their prominence; the three most prominent are taken as the three bright fringes, labelled $1, 2, 3$ in order of increasing x.

3. **Valleys.** For peak k the valley is the position of the lowest value of the smoothed curve between peak $k-1$ and peak k (for $k = 1$: between the start of the profile and peak 1).

4. **Refinement.** The luma values are read from a lightly smoothed curve (Gaussian, $\sigma = 1.5$ pixels). The peak is placed at the highest, and the valley at the lowest, point of this curve within $\pm\sigma_d$ of the positions found in steps 2–3. This avoids a single noisy pixel setting $L_{\max}$ or $L_{\min}$.

5. **Visibility.** V_1, V_2, V_3 follow from Eq. (22) and V_{avg} from Eq. (23).

6. **Quality flag.** A photograph is *flagged* when, for any of its three pairs, the peak and its valley are ≤ 7 pixels apart. In that case the fringe is not resolved (the smoothing scale is comparable to the peak–valley spacing) and V is unreliable. Flagged photographs: 3, 4, 25, 39 (at $z = 0.4$, 0.6, 4.8, 8.8 mm). They are kept in all main results; fits without them are given for comparison.

6.1 Photographic dataset and individual estimates

Figure 8 collects the peak-intensity graphs for all 42 photographs. Each plot is labeled with its sheet number and the three pair visibilities and average visibility. The exact peak and valley positions and luma values are listed in Table 8; the mirror displacements and visibility estimates with uncertainties are reported in Table 7. The all-data trend, unresolved-profile sensitivity and residual diagnostics are shown in Figs. 9–11.

Peak-intensity profiles for all 42 photographs

Each graph marks the selected bright peaks and dark valleys; its title reports V_1, V_2, V_3 and their mean.

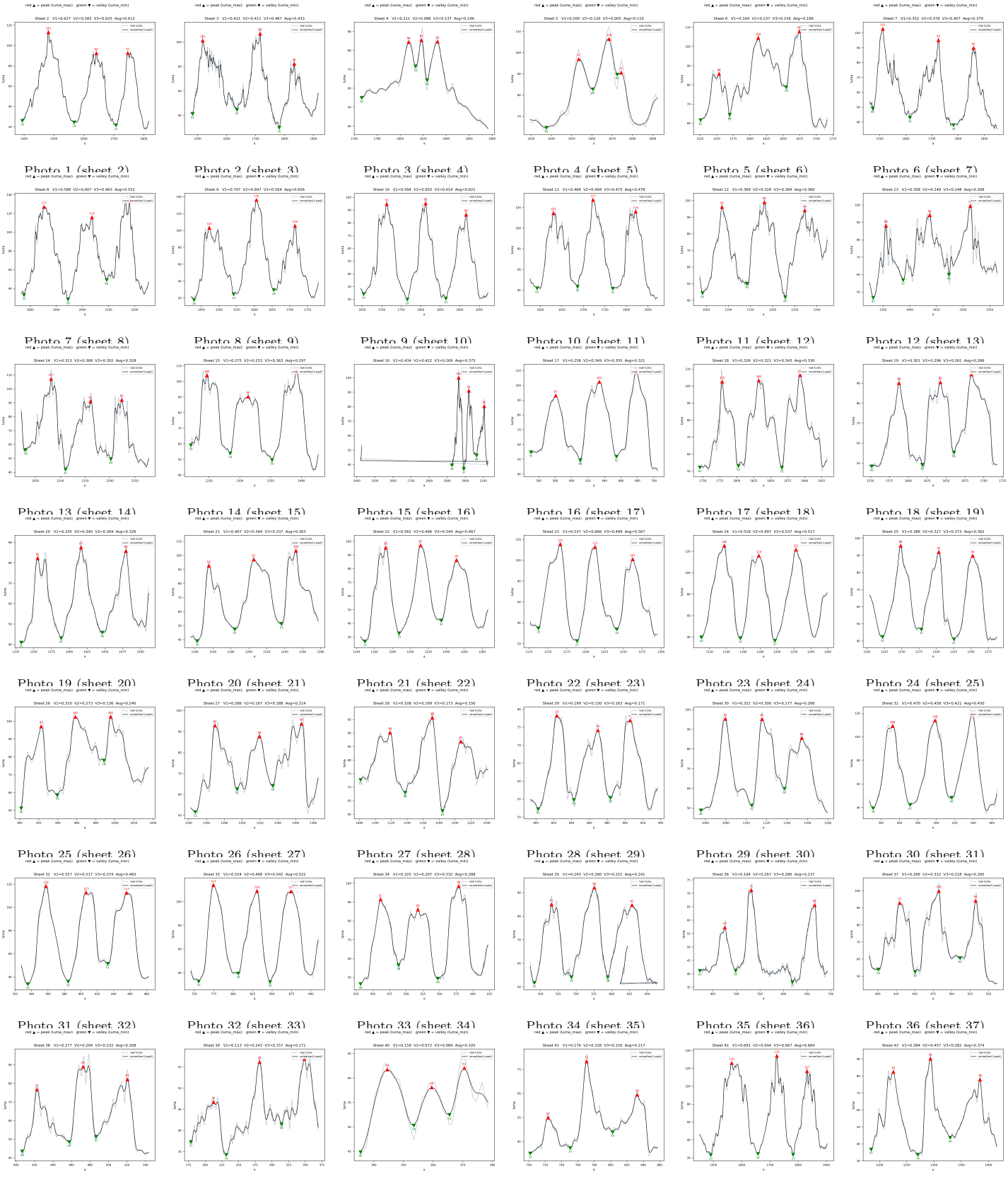


Figure 8: Peak-intensity graphs for photographs 1–42 (source sheets 2–43). Numerical peak and valley readings, displacements, and visibility estimates are given in the results tables.

Table 7: Visibilities and uncertainties for all photographs. δ_{prop}: noise propagation; δ_{scat}: scatter of the three pairs ($s/\sqrt{3}$); δ_{meth}: sensitivity to analysis settings; δ_{tot}: quadrature sum (Section 6.3). ∘ = flagged photograph.

Photo	Sheet	z (mm)	V_1	V_2	V_3	V_{avg}	δ_{prop}	δ_{scat}	δ_{meth}	δ_{tot}	Flag
1	2	0.0	0.627	0.583	0.625	0.612	0.009	0.014	0.009	0.019	
2	3	0.2	0.421	0.411	0.467	0.433	0.014	0.017	0.014	0.026	
3	4	0.4	0.211	0.088	0.137	0.146	0.009	0.036	0.033	0.050	∘
4	5	0.6	0.200	0.126	0.005	0.110	0.011	0.057	0.032	0.066	∘
5	6	0.8	0.164	0.237	0.156	0.186	0.006	0.026	0.011	0.029	
6	7	1.0	0.352	0.378	0.407	0.379	0.006	0.016	0.009	0.019	
7	8	1.2	0.588	0.607	0.463	0.552	0.008	0.045	0.010	0.047	
8	9	1.4	0.707	0.697	0.564	0.656	0.013	0.046	0.011	0.049	
9	10	1.6	0.594	0.655	0.614	0.621	0.011	0.018	0.014	0.025	
10	11	1.8	0.466	0.494	0.475	0.478	0.012	0.008	0.010	0.017	
11	12	2.0	0.369	0.328	0.384	0.360	0.008	0.017	0.012	0.022	
12	13	2.2	0.308	0.248	0.248	0.268	0.013	0.020	0.022	0.033	
13	14	2.4	0.313	0.368	0.302	0.328	0.014	0.021	0.016	0.030	
14	15	2.6	0.275	0.253	0.363	0.297	0.009	0.034	0.010	0.036	
15	16	2.8	0.434	0.422	0.269	0.375	0.012	0.053	0.019	0.058	
16	17	3.0	0.258	0.349	0.355	0.321	0.008	0.031	0.013	0.035	
17	18	3.2	0.326	0.321	0.343	0.330	0.011	0.007	0.019	0.023	
18	19	3.4	0.301	0.296	0.261	0.286	0.009	0.013	0.016	0.022	
19	20	3.6	0.335	0.340	0.304	0.326	0.009	0.011	0.013	0.020	
20	21	3.8	0.407	0.344	0.337	0.363	0.012	0.022	0.018	0.031	
21	22	4.0	0.562	0.496	0.344	0.467	0.013	0.065	0.018	0.068	
22	23	4.2	0.537	0.666	0.499	0.567	0.011	0.051	0.017	0.055	
23	24	4.4	0.518	0.497	0.537	0.517	0.013	0.012	0.020	0.026	
24	25	4.6	0.386	0.327	0.373	0.362	0.008	0.018	0.013	0.024	
25	26	4.8	0.310	0.273	0.136	0.240	0.010	0.053	0.019	0.057	∘
26	27	5.0	0.286	0.167	0.188	0.214	0.010	0.037	0.015	0.041	
27	28	5.2	0.108	0.169	0.173	0.150	0.005	0.021	0.008	0.023	
28	29	5.4	0.199	0.150	0.163	0.171	0.006	0.015	0.009	0.018	
29	30	5.6	0.322	0.300	0.177	0.266	0.009	0.045	0.019	0.050	
30	31	5.8	0.470	0.458	0.421	0.450	0.012	0.015	0.017	0.025	
31	32	6.0	0.557	0.517	0.374	0.483	0.010	0.055	0.015	0.058	
32	33	6.2	0.554	0.468	0.542	0.522	0.009	0.027	0.015	0.032	
33	34	6.4	0.325	0.207	0.332	0.288	0.006	0.041	0.008	0.042	
34	35	6.8	0.243	0.260	0.221	0.241	0.007	0.011	0.012	0.018	
35	36	7.2	0.164	0.267	0.280	0.237	0.007	0.037	0.007	0.038	
36	37	7.6	0.266	0.312	0.218	0.265	0.010	0.027	0.012	0.032	
37	38	8.0	0.277	0.294	0.232	0.268	0.014	0.019	0.019	0.030	
38	39	8.4	0.113	0.243	0.157	0.171	0.007	0.038	0.009	0.040	
39	40	8.8	0.159	0.072	0.084	0.105	0.010	0.027	0.022	0.036	∘
40	41	9.2	0.176	0.326	0.150	0.217	0.007	0.055	0.012	0.057	
41	42	9.6	0.691	0.694	0.667	0.684	0.010	0.008	0.014	0.019	
42	43	10.0	0.384	0.457	0.282	0.374	0.009	0.051	0.013	0.053	

Summary of the measured values

- Number of photographs: 42; displacement range 0 to 10.0 mm.
- V_{avg}: minimum 0.105, maximum 0.684, mean 0.350, standard deviation 0.153.
- Mean absolute difference of V_{avg} between neighbouring photographs: 0.101.

6.2 Regression and sensitivity analyses

6.2.1 Worked example: photograph 8 ($z = 1.4$ mm)

Photograph 8 is sheet 9 of the workbook (361 profile points). The three peak–valley pairs found by the procedure of Section 6 are:

Pair	x_{peak} (px)	$L_{\max}$	x_{valley} (px)	$L_{\min}$
1	1472.5	103.17	1429.5	17.72
2	1605.5	136.18	1541.5	24.29
3	1714.5	105.55	1654.5	29.40

From Eq. (22):

$$V_1 = \frac{103.17 - 17.72}{103.17 + 17.72} = \frac{85.45}{120.89} = 0.7068,$$
$$V_2 = \frac{136.18 - 24.29}{136.18 + 24.29} = \frac{111.89}{160.47} = 0.6973,$$
$$V_3 = \frac{105.55 - 29.40}{105.55 + 29.40} = \frac{76.15}{134.95} = 0.5643.$$

The average visibility, Eq. (23), is

$$V_{\text{avg}} = \frac{0.7068 + 0.6973 + 0.5643}{3} = 0.6561.$$

Uncertainty of this photograph. The pixel noise on this profile (RMS difference between the raw and lightly smoothed luma) is $\sigma_L = 1.78$ luma units, which is taken as $\delta L_{\max} = \delta L_{\min}$ (Section 6.3). For pair 1, Eq. (25) gives

$$\delta V_1 = \sqrt{\left(\frac{2\,17.72}{120.89^2}\right)^2 1.78^2 + \left(\frac{2\,103.17}{120.89^2}\right)^2 1.78^2}$$
$$= \sqrt{(0.00243 \times 1.78)^2 + (0.01412 \times 1.78)^2} = 0.0255,$$

and in the same way $\delta V_2 = 0.0191$, $\delta V_3 = 0.0214$. Equation (26) gives

$$\delta V_{\text{avg}}^{\text{prop}} = \tfrac{1}{3}\sqrt{6.485 + 3.647 + 4.575} \times 10^{-2} = \tfrac{1}{3}\sqrt{14.707 \times 10^{-4}} = 0.0128.$$

The scatter of the three pair values about their mean gives $s = 0.0797$ and $\delta_{\text{scat}} = s/\sqrt{3} = 0.0460$. Repeating the analysis with six different smoothing settings gives V_{avg} between 0.6357 and 0.6561, a standard deviation of $\delta_{\text{meth}} = 0.0111$. The total is

$$\delta_{\text{tot}} = \sqrt{0.0128^2 + 0.0460^2 + 0.0111^2} = 0.0490,$$

so for photograph 8, $V_{\text{avg}} = 0.6561 \pm 0.0490$.

6.2.2 Linear fit of V_{avg} against z (all 42 photographs)

The sums needed for the least-squares fit are:

$$n = 42, \quad \sum z = 181.2 \text{ mm}, \quad \sum V = 14.6877,$$
$$\sum z^2 = 1102.24 \text{ mm}^2, \quad \sum zV = 60.0066, \quad \sum V^2 = 6.1007.$$

The means are $\bar{z} = 4.3143$ mm and $\bar{V} = 0.350$. Then

$$S_{zz} = \sum z^2 - n\bar{z}^2 = 320.491 \text{ mm}^2,$$
$$S_{zV} = \sum zV - n\bar{z}\bar{V} = -3.3605 \text{ mm},$$
$$S_{VV} = 0.9643.$$

Slope and intercept:

$$b = \frac{S_{zV}}{S_{zz}} = \frac{-3.3605}{320.491} = -0.01049 \text{ mm}^{-1},$$
$$a = \bar{V} - b\bar{z} = 0.350 - (-0.01049)(4.3143) = 0.395.$$

The residual sum of squares is $\sum e_i^2 = 0.9290$, so the residual standard deviation is

$$s = \sqrt{\frac{0.9290}{42-2}} = 0.152.$$

Uncertainties:

$$\delta b = \frac{s}{\sqrt{S_{zz}}} = \frac{0.152}{\sqrt{320.491}} = 0.00851 \text{ mm}^{-1},$$
$$\delta a = s\sqrt{\frac{1}{n} + \frac{\bar{z}^2}{S_{zz}}} = 0.044.$$

Goodness of fit and significance:

$$r = \frac{S_{zV}}{\sqrt{S_{zz}S_{VV}}} = -0.19, \qquad R^2 = 0.037, \qquad R^2_{\text{adj}} = 0.012,$$
$$t = \frac{b}{\delta b} = -1.23, \qquad F = t^2 = 1.52, \qquad p = 0.23 \ \ (\text{d.o.f.} = 42-2 = 40).$$

With $t_{0.975,40} = 2.021$ the 95% confidence intervals are

$$b = -0.01049 \pm 2.021 \times 0.00851: \ \ [-0.0277,\ 0.0067] \text{ mm}^{-1}, \qquad a: \ \ [0.307,\ 0.483].$$

The result is

$$\boxed{V_{\text{avg}} = (0.395 \pm 0.044) + (-0.01049 \pm 0.00851)\, z} \qquad (z \text{ in mm}). \tag{27}$$

The fitted value falls from 0.395 at $z = 0$ to 0.290 at $z = 10$ mm, a total fall of 0.105 over the full range.

6.2.3 Fit without the flagged photographs

Repeating the calculation with the 38 unflagged photographs gives

$$V_{\text{avg}} = (0.435 \pm 0.044) + (-0.01460 \pm 0.00859)\, z,$$
$$r = -0.27,\ R^2 = 0.074,\ s = 0.141,\ t = -1.70,\ p = 0.10,$$

with a 95% confidence interval for the slope of $[-0.0320,\ 0.0028]$ mm^{-1}.

6.2.4 Block means

To see the trend without assuming a straight line, the photographs were grouped by displacement:

z range (mm)	photographs	mean V_{avg}	standard deviation
0.0–2.5	13	0.395	0.185
2.5–5.0	12	0.371	0.098
5.0–7.5	10	0.302	0.133
7.5–10.0	7	0.298	0.190

6.3 Uncertainty and model diagnostics

The visibility of a photograph is affected by random errors (pixel noise, speckle), by the choice of the analysis procedure, and by systematic effects of the recording. The following subsections quantify each.

6.3.1 Sources of uncertainty in V_{avg}

Source	Origin	Treatment
Pixel noise & speckle	Random fluctuations of luma along the profile	δ_{prop}, Eqs. (25),(26)
Pair-to-pair scatter	The three fringes of one profile have genuinely different contrast	$\delta_{\text{scat}} = s/\sqrt{3}$
Analysis settings	Smoothing widths change which points are chosen as peak/valley	δ_{meth}
Mirror position z	Reading of the coarse knob	Section 6.3.6
Gamma / exposure	Luma is not linear in intensity; exposure unknown	Section 6.3.11
Framing & alignment	Pattern size, tilt and position change between photographs	Discussed (not quantifiable)

6.3.2 Pixel noise and propagation

For every profile the noise level σ_L is estimated as the RMS difference between the raw luma and the lightly smoothed luma over the whole profile:

$$\sigma_L = \sqrt{\frac{1}{N_p}\sum_x [y(x) - L(x)]^2}.$$

Over the 42 photographs σ_L lies between 0.80 and 2.37 luma units (mean 1.55). This is a conservative estimate for the uncertainty of one extremum, because the extremum is read from the smoothed curve, which has less noise than the raw pixels. It was used as $\delta L_{\max} = \delta L_{\min} = \sigma_L$ in Eq. (25) for each pair, and the three pair uncertainties were combined with Eq. (26). The mean value of δ_{prop} is 0.010.

A limitation: choosing the highest point of a noisy curve as the peak and the lowest as the valley tends to overestimate $L_{\max} - L_{\min}$ (a small positive bias in V). This bias is not included in δ_{prop}; it is reduced by the smoothing, and its size is indicated by δ_{meth}.

6.3.3 Scatter between the three pairs

The three pair visibilities of one photograph differ by more than the pixel noise would explain, because the fringes at different positions in the beam really have different contrast (the beam intensity varies across the screen and the fringe pattern is not uniform). The standard error of the mean of the three values is

$$\delta_{\text{scat}} = \frac{1}{\sqrt{3}}\sqrt{\frac{\sum_k (V_k - V_{\text{avg}})^2}{3-1}}.$$

Its mean value is 0.029, about three times δ_{prop}, so *the variation between fringes is the largest contribution.* With only three values per photograph this quantity is itself uncertain.

6.3.4 Sensitivity to the analysis settings

The analysis was repeated for six combinations of smoothing parameters: detection width $\sigma_d \in \{1.5\%, 2.5\%, 4\%\}$ of the profile length and luma smoothing $\sigma \in \{1.5, 3.0\}$ pixels. The standard deviation of the six values of V_{avg} for each photograph is δ_{meth} (mean 0.015). The valley convention was also tested: taking the valley *after* each peak instead of before gives a fitted slope of -0.01328 mm^{-1} ($r = -0.24$, $p = 0.13$), compared with -0.01049 for the adopted convention. The sign and approximate size of the trend do not depend on which convention is used, and neither is significant.

6.3.5 Total uncertainty of V_{avg}

The three contributions are independent in origin and are added in quadrature:

$$\delta_{\text{tot}} = \sqrt{\delta_{\text{prop}}^2 + \delta_{\text{scat}}^2 + \delta_{\text{meth}}^2}.$$

Mean $\delta_{\text{tot}} = 0.036$, with individual values from 0.017 to 0.068 (Table 7). The error bars in Fig. 9 are δ_{tot}. This uncertainty describes how well V_{avg} is determined *from the profile that was extracted*; it does not include changes of framing, tilt or exposure between photographs.

6.3.6 Uncertainty in z

The mirror displacement is the difference of two coarse-knob readings, each with a reading uncertainty of about half the least count, 0.005 mm. Hence $\delta z \approx \sqrt{2} \times 0.005 \approx 0.007$ mm. This is much smaller than the step between photographs (0.2 or 0.4 mm) and than the width of the features in Fig. 9. Its effect on the fitted slope is of order $|b|\,\delta z/\Delta z \ll \delta b$ and is negligible. Backlash, if present, could cause a systematic offset between the knob reading and the true mirror position; this was avoided by turning the knob in one direction only.

6.3.7 Uncertainty of the fit parameters

For the fit to all 42 photographs (Section 6.2.2):

Quantity	Value	Uncertainty
Slope b (mm^{-1})	-0.01049	0.00851 (95% CI: -0.0277 to 0.0067)
Intercept a	0.395	0.044 (95% CI: 0.307 to 0.483)
Residual standard deviation s	0.152	—

The 95% confidence interval for the slope contains zero, so a flat line (no dependence on z) is consistent with the data.

6.3.8 Weighted fit and chi-square

A weighted fit using δ_{tot} as the uncertainty of each point gives slope -0.00945 mm^{-1} and intercept 0.405, with a formal slope uncertainty of 0.00162 mm^{-1}. The goodness of fit is, however, very poor: $\chi^2 = 1193$ for 40 degrees of freedom, a reduced chi-square of $\chi^2_\nu = 29.8$. A straight line is therefore *not* a good description of the data within the stated measurement uncertainties; the points scatter around the line by about $\sqrt{\chi^2_\nu} \approx 5.5$ times their error bars. This means that the variations of V_{avg} from photograph to photograph are real changes (or changes caused by effects not included in δ_{tot}), not measurement noise. Because of this the formal weighted uncertainty is far too small; scaling it by $\sqrt{\chi^2_\nu}$ gives 0.00882 mm^{-1}, which agrees with the unweighted value 0.00851. The unweighted fit is therefore the appropriate summary and is the one quoted as the main result.

6.3.9 Residual diagnostics and effective sample size

The residuals of the main fit are plotted in Fig. 11. The Shapiro–Wilk test gives $W = 0.969$, $p = 0.30$, so the residuals are consistent with a normal distribution. However, the residuals are *not* independent: the Durbin–Watson statistic is $DW = 0.83$ (about 2 expected) and the lag-1 autocorrelation is $\rho = 0.57$. Neighbouring photographs deviate from the line in the same direction, which is what is expected if V_{avg} varies in broad waves rather than randomly. The effective number of independent points is $n_{\text{eff}} \approx 11.5$, so the slope uncertainty from the ordinary formula is too small by a factor $\sqrt{n/n_{\text{eff}}} \approx 1.9$: a corrected value is $\delta b \approx 0.0163$ mm^{-1} (corresponding to $p \approx 0.54$). The conclusion that the slope is not significant is therefore strengthened.

6.3.10 Robustness tests of the slope

Test	Result	Comment
Ordinary least squares, all 42	-0.01049 ± 0.00851	$p = 0.23$
OLS, 38 unflagged	-0.01460 ± 0.00859	$p = 0.10$
Weighted least squares (δ_{tot})	-0.00945 ± 0.00882	uncertainty scaled by $\sqrt{\chi^2_\nu}$
Theil–Sen (median slope)	-0.01626	95% CI -0.0315 to 0.0032
Bootstrap (10 000 resamples)	-0.01092 ± 0.01056	percentile 95% CI -0.0315 to 0.0099; 85% of slopes negative
Permutation test	—	$p = 0.23$
Spearman rank correlation	$\rho_s = -0.23$	$p = 0.15$
Kendall rank correlation	$\tau = -0.18$	$p = 0.10$
Leave-one-out slopes	-0.01772 to -0.00731	extremes on omitting photographs 41 and 1

All methods agree on a small negative slope of order -0.01 mm^{-1}. No method makes it significant at the 5% level, and in the bootstrap and leave-one-out tests the sign is stable but the interval always includes zero. No single photograph controls the result: omitting any one changes the slope only between -0.01772 and -0.00731.

6.3.11 Gamma encoding of the luma

The visibilities above are computed from luma values. If the luma is converted back to linear intensity using Eq. (20) with $\gamma_c = 2.2$ (the usual value for sRGB images; the true processing of this camera is not known), the same peaks and valleys give a mean visibility of 0.624 instead of 0.350, so the absolute level of V depends strongly on this assumption. The two sets of values are, however, almost perfectly correlated (correlation coefficient 0.979) and the linear fit of the linearised visibility gives a slope of -0.0144 ± 0.0111 mm^{-1} ($r = -0.20$, $p = 0.20$). So the *trend* (small, negative, not significant) is unchanged; only the absolute scale of V is uncertain. The values in this report are best read as a relative measure of contrast between photographs.

6.3.12 Peak and valley luma separately

The mean peak luma $\langle L_{\max} \rangle$ and the mean valley luma $\langle L_{\min} \rangle$ of each photograph was also fitted against z. The peak luma decreases with z (slope -2.26 ± 0.85 luma/mm, $r = -0.39$, $p = 0.01$) while the valley luma shows no significant trend (slope -0.50 ± 0.71 luma/mm, $p = 0.48$). A fall of visibility caused by a loss of coherence would be expected to raise $L_{\min}$ relative to $L_{\max}$. Here the small fall in V_{avg} goes with a fall in the brightness of the bright fringes, which could equally come from the overall brightness of the pattern (exposure, beam alignment, framing), which was not controlled. This again shows that the data do not identify the cause.

6.3.13 Overall statement of the uncertainty

Combining the above:

- Each value of V_{avg} is known to about ± 0.036 (typical δ_{tot}), dominated by the difference between fringes of the same photograph.
- The photograph-to-photograph variation (standard deviation 0.153, neighbour difference 0.101) is much larger than this and is therefore real.
- The linear slope with z is -0.01049 ± 0.00851 mm^{-1} (or ± 0.0163 allowing for correlated residuals), not significantly different from zero.
- The absolute scale of V depends on the gamma encoding; the trend does not.

6.4 Results and interpretation

6.4.1 Contrast plots and diagnostics

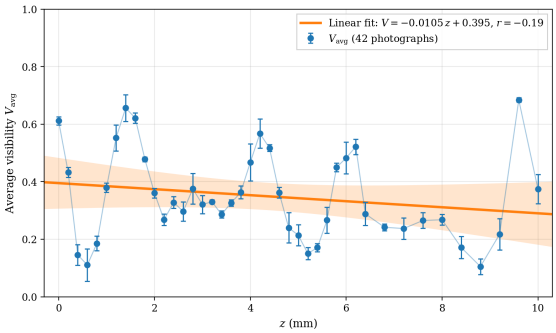


Figure 9: Average luma visibility against mirror displacement for all 42 photographs. Error bars show total within-image uncertainty; the solid line is the all-data least-squares fit.

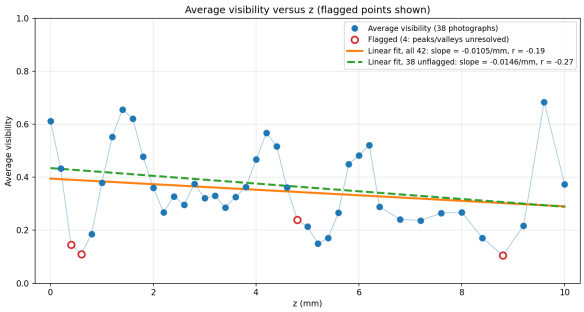


Figure 10: Sensitivity to the four unresolved profiles: all-data and 38-image fits are shown together.

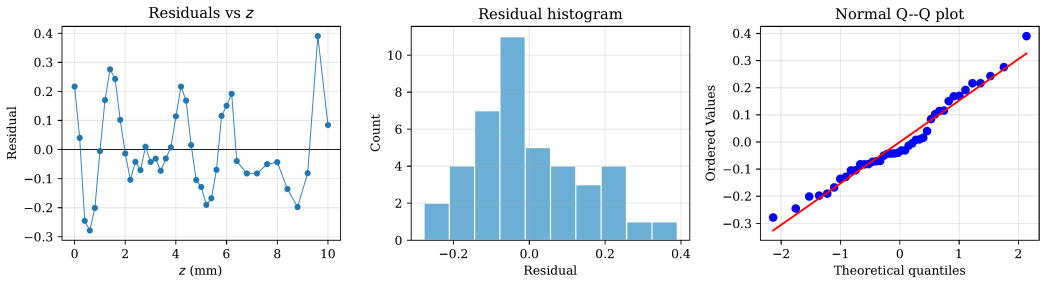


Figure 11: Residual diagnostics for the all-data fit; adjacent residuals show structure and serial correlation.

6.4.2 Summary of contrast results

Quantity	Value
Number of photographs	42
Displacement range	0 to 10.0 mm
Mean V_{avg} (std. dev.)	0.350 (0.153)
Range of V_{avg}	0.105 to 0.684
Part 2 wavelength	533.3 nm (stated source value)
Visibility at $z = 10.0$ mm (Photo 42)	0.374
Operational coherence range	$\gtrsim$ 20 mm path difference, conditional on $z = 0$ being near equal-arm position
Typical uncertainty of one V_{avg}	± 0.036
Fit, all points: slope	-0.01049 ± 0.00851 mm^{-1}
Fit, all points: intercept	0.395 ± 0.044
r, R^2, R^2_{adj}	-0.19, 0.037, 0.012
p-value of the slope	0.23
95% CI of the slope	$[-0.0277, 0.0067]$ mm^{-1}
Residual standard deviation	0.152
Fit, 38 unflagged: slope, r, p	-0.01460 mm^{-1}, -0.27, 0.10

6.4.3 Interpretation of the contrast trend

6.4.4 How does the visibility change with z?

The fitted line falls from 0.395 at $z = 0$ to 0.290 at $z = 10$ mm, so on average the visibility is lower for larger mirror displacements. The block means in Section 6.2.4 also decrease (from about 0.39 in the first block to about 0.30 in the last). However, the fit is poor: $R^2 = 0.037$ means that only about 4% of the variation of V_{avg} is explained by z, the residual standard deviation (0.152) is almost equal to the standard deviation of V_{avg} itself (0.153), and the 95% confidence interval of the slope includes zero. The fitted fall over the whole range (0.105) is about the same as the typical difference between two neighbouring photographs (0.101). At the end of the scan (Photo 42, $z = 10.0$ mm), the measured luma visibility is 0.374, so fringes remain measurable after a $2z = 20.0$ mm change in optical path difference. If the initial position was close to equal arm lengths and the recorded contrast is taken as evidence of coherence, this gives an operational lower-bound scale $\ell_c \gtrsim 20$ mm. It is conditional rather than a fitted coherence length: the absolute zero-path-difference position was not recorded, the luma visibility is not calibrated photometric visibility, and the visibility trend is not statistically significant. With $\lambda = 533.3$ nm, Eq. (19) would associate this scale with a linewidth of order $\Delta\lambda \lesssim \lambda^2/\ell_c \approx 0.014$ nm, but the present data do not measure that linewidth.

6.4.5 Structure not captured by a straight line

Fig. 9 shows that V_{avg} rises and falls in waves, with clear maxima near $z \approx 1.4$, 4.2, 6.2 and 9.6 mm and minima in between (near 0.6, 2.2, 5.2 and 8.8 mm); the first photograph ($z = 0$) is also high. The positive autocorrelation of the residuals (lag-1 correlation 0.57, Durbin–Watson 0.83) confirms that this is systematic structure and not random noise. The source is treated as a single spectral component; no two-line beat period is predicted. The observed waves in Fig. 9, with spacing of order 2–3 mm, are not sufficient to determine a spectral beat or a coherence envelope, particularly because exposure, alignment and other recording conditions were not held fixed.

6.4.6 Flagged photographs

Photographs 3, 4, 25, 39 are flagged because the fringes were not resolved well enough by the analysis. Removing them steepens the slope slightly (-0.01460 instead of -0.01049 mm^{-1}) and increases $|r|$ from -0.19 to -0.27, but the slope is still not significant ($p = 0.10$). They do not change the conclusion, and they are kept in the main result for completeness.

6.4.7 Physical interpretation

The data show a weak, non-significant average decrease of visibility with mirror displacement, superimposed on strong photograph-to-photograph variations. By Eq. (18) the visibility equals the degree of temporal coherence $|\gamma(\tau)|$ at the delay $\tau = 2z/c$, so changes of V with z reflect the coherence of the source, but also anything else that changes the contrast of the recorded pattern. The following explanations are possible; the experiment does not allow us to decide between them.

- **Finite coherence length.** A finite spectral width causes $|\gamma|$ to decay with path difference. The measured visibility remains appreciable at the largest recorded displacement, but the large scatter and changing image conditions prevent a reliable fit of the coherence envelope or a precise intrinsic coherence length.
- **Changes of alignment.** Moving the mirror with the coarse knob can slightly tilt M_1 or change the overlap of the two beams, which changes the intensity ratio I_1/I_2 and the fringe shape and therefore V through $V = 2\sqrt{I_1 I_2}\,|\gamma|/(I_1 + I_2)$. The tilt was not recorded.
- **Recording conditions.** Exposure, camera position, framing, background light and speckle may differ between photographs. Because luma is a non-linear function of intensity (Eq. (20)), changes of exposure change V_L even if the true contrast is constant.

- **Analysis effects.** Visibility is measured along a single line; different parts of the pattern have different contrast, and the choice of line and of the three fringes changes the result (shown by δ_{scat} and δ_{meth}).

Because several things changed or were not recorded together with z, the experiment does not show that z alone caused any change in visibility.

Peak–valley readings for all photographs

The table gives the peak and preceding-valley positions and luma values used for each photograph's fringe visibility, plus the profile noise estimate σ_L.

Table 8: Positions x (pixels) and luma values of the peaks and of the valleys preceding them, and the noise level σ_L of each profile.

Photo	z	x_{p1}	$L^{(1)}_{\max}$	x_{v1}	$L^{(1)}_{\min}$	x_{p2}	$L^{(2)}_{\max}$	x_{v2}	$L^{(2)}_{\min}$	x_{p3}	$L^{(3)}_{\max}$	x_{v3}	$L^{(3)}_{\min}$	σ_L
1	0.0	1481.5	113.02	1394.5	25.92	1641.5	92.44	1567.5	24.33	1746.5	92.62	1707.5	21.39	1.19
2	0.2	1517.5	101.23	1482.5	41.22	1715.5	106.81	1635.5	44.60	1833.5	82.16	1781.5	29.88	2.00
3	0.4	1807.5	84.39	1766.5	54.93	1818.5	85.42	1813.5	71.54	1832.5	84.62	1823.5	64.24	1.65
4	0.6	1653.5	96.95	1637.5	64.65	1668.5	106.42	1660.5	82.68	1674.5	90.62	1672.5	89.63	2.37
5	0.8	1553.5	85.73	1525.5	61.58	1612.5	104.41	1569.5	64.40	1674.5	107.88	1654.5	78.78	1.20
6	1.0	1511.5	102.72	1472.5	49.26	1724.5	95.11	1615.5	42.89	1858.5	89.84	1782.5	37.82	0.98
7	1.2	1852.5	126.92	1776.5	32.96	2036.5	115.76	1944.5	28.33	2179.5	133.57	2092.5	49.04	1.43
8	1.4	1472.5	103.17	1429.5	17.72	1605.5	136.18	1541.5	24.29	1714.5	105.55	1654.5	29.40	1.78
9	1.6	1712.5	94.64	1653.5	24.09	1812.5	95.21	1765.5	19.83	1916.5	86.35	1864.5	20.66	1.27
10	1.8	1630.5	114.52	1592.5	41.68	1722.5	127.29	1686.5	43.14	1821.5	116.03	1768.5	41.33	2.09
11	2.0	1085.5	96.12	1041.5	44.30	1181.5	98.96	1141.5	50.05	1272.5	94.06	1228.5	41.89	1.28
12	2.2	1355.5	88.07	1331.5	46.61	1437.5	94.13	1387.5	56.76	1513.5	99.43	1473.5	59.91	2.35
13	2.4	2081.5	107.13	2030.5	56.04	2160.5	90.87	2110.5	41.96	2223.5	92.06	2201.5	49.40	2.37
14	2.6	1246.5	103.90	1220.5	59.05	1313.5	90.26	1284.5	53.84	1392.5	105.86	1352.5	49.52	1.62
15	2.8	1963.5	99.99	1926.5	39.45	2019.5	90.97	1991.5	36.95	2104.5	80.32	2062.5	46.25	1.74
16	3.0	580.5	92.98	550.5	54.80	633.5	102.36	610.5	49.39	678.5	108.83	654.5	51.79	1.36
17	3.2	1778.5	102.57	1745.5	52.14	1832.5	103.23	1802.5	53.09	1893.5	106.45	1866.5	52.02	2.04
18	3.4	1587.5	90.07	1551.5	48.35	1642.5	90.48	1618.5	49.18	1683.5	94.37	1660.5	55.26	1.50
19	3.6	1355.5	82.18	1332.5	40.92	1416.5	87.46	1388.5	43.08	1479.5	85.77	1446.5	45.80	1.34
20	3.8	1155.5	92.56	1142.5	38.98	1205.5	97.00	1184.5	47.32	1252.5	103.32	1236.5	51.25	1.90
21	4.0	1172.5	95.30	1149.5	26.71	1211.5	96.96	1187.5	32.68	1251.5	86.13	1234.5	42.04	1.77
22	4.2	1166.5	115.34	1137.5	34.73	1212.5	112.70	1188.5	22.56	1262.5	100.86	1241.5	33.75	1.67
23	4.4	1137.5	125.06	1110.5	39.74	1178.5	115.84	1156.5	38.97	1222.5	121.54	1197.5	36.60	2.21
24	4.6	1248.5	95.55	1222.5	42.28	1303.5	91.90	1278.5	46.65	1352.5	89.71	1325.5	40.92	1.18
25	4.8	922.5	97.09	901.5	51.12	958.5	102.45	939.5	58.55	995.5	102.49	988.5	77.88	1.87
26	5.0	1269.5	92.87	1247.5	51.53	1319.5	87.60	1294.5	62.47	1366.5	93.71	1334.5	64.07	1.78
27	5.2	1118.5	90.17	1081.5	72.64	1171.5	95.64	1137.5	67.92	1207.5	86.86	1184.5	61.21	1.01
28	5.4	823.5	78.17	802.5	52.22	869.5	74.12	842.5	54.73	905.5	76.90	883.5	55.34	0.91
29	5.6	1079.5	95.30	1055.5	48.83	1115.5	95.14	1105.5	51.26	1154.5	85.50	1137.5	59.75	1.50
30	5.8	552.5	109.10	531.5	39.35	598.5	113.80	571.5	42.26	638.5	118.60	616.5	48.27	2.07
31	6.0	357.5	118.27	335.5	33.68	406.5	112.79	384.5	35.90	455.5	112.58	432.5	51.27	1.67
32	6.2	774.5	113.77	755.5	32.69	830.5	108.73	806.5	39.36	874.5	108.59	847.5	32.21	1.43
33	6.4	461.5	91.32	431.5	46.48	517.5	86.00	488.5	56.47	578.5	98.46	547.5	49.37	1.03
34	6.8	515.5	84.97	491.5	51.71	575.5	92.10	543.5	54.09	628.5	84.70	594.5	54.07	1.18
35	7.2	475.5	57.22	422.5	41.08	531.5	71.16	498.5	41.18	665.5	65.61	618.5	36.87	0.83
36	7.6	429.5	92.84	400.5	53.84	482.5	99.90	450.5	52.44	532.5	94.14	511.5	60.50	1.87
37	8.0	422.5	76.69	406.5	43.42	472.5	88.97	457.5	48.50	520.5	82.20	486.5	51.29	2.09
38	8.4	212.5	46.70	178.5	37.25	281.5	56.19	231.5	34.20	348.5	56.94	314.5	41.46	0.80
39	8.8	344.5	61.74	335.5	44.83	359.5	58.08	353.5	50.26	370.5	62.05	365.5	52.44	1.31
40	9.2	723.5	49.90	701.5	35.00	770.5	73.27	750.5	37.27	832.5	59.42	802.5	43.91	0.88
41	9.6	1564.5	125.98	1490.5	23.00	1724.5	133.89	1657.5	24.15	1831.5	116.85	1782.5	23.30	1.51
42	10.0	1225.5	82.44	1184.5	36.67	1293.5	90.22	1270.5	33.62	1384.5	77.88	1329.5	43.64	1.18

7 Synthesis of the two measurements

The two measurements use the same instrument but address different quantities: the wavelength measurement determines λ from fringe counts, while the fringe-visibility analysis measures contrast (V) from photographs. The results are independent; their connections are described below.

7.1 Comparison of the two experiments

	Wavelength measurement	Fringe visibility
Source	He–Ne laser (632.8 nm)	Single Part 2 source (533.3 nm)
Fringes	concentric circles (parallel film)	circular, elliptical or curved (see photographs)
Quantity measured	N fringes per displacement $\Rightarrow \lambda$	V from luma profile $\Rightarrow V_{\text{avg}}(z)$
Mirror travel	0.100 mm in 10 steps of 0.010 mm	0 to 10.0 mm in steps of 0.2 and 0.4 mm
Recording	counted by eye at the centre	photographs, one luma profile per photograph
Main result	$\lambda = 645.2 \pm 3.1$ nm (1.95% above 632.8 nm)	slope -0.01049 ± 0.00851 mm^{-1}, not significant

7.2 Wavelength measurement and the fringe scale

The wavelength measurement shows that one fringe corresponds to a mirror displacement of $\lambda/2$ (measured: 322.6 nm for the He–Ne line; expected 316.4 nm). For the Part 2 source, $\lambda/2 = 266.65$ nm. The step between two visibility photographs is therefore very large compared with the fringe scale:

$$\frac{0.2\text{ mm}}{266.65\text{ nm}} \approx 750 \text{ fringes per 0.2 mm step,} \qquad \frac{0.4\text{ mm}}{266.65\text{ nm}} \approx 1500 \text{ fringes per 0.4 mm step,}$$

and the whole range of 10.0 mm corresponds to about 37 502 fringes. In the laser measurement, each 0.010 mm step produced only about 31 fringes. The mirror displacement between visibility photographs is therefore about 20 times larger, and the controlled parameter is the mirror displacement z, not the photograph number. No two-line beat calculation is applicable to the stated single-source Part 2 setup.

7.3 Effect of wavelength calibration on the visibility analysis

In the He–Ne measurement, 310 fringes were counted where 316.1 were expected over the nominal 0.100 mm. If this shortfall were entirely due to the coarse scale (a systematic calibration effect that these data cannot separate from under-counting and backlash), the true mirror displacement would be $f = 310/316.1 = 0.9807$ times the nominal displacement. The z values in the Part 2 visibility series would then be too large by about 2%, and the slope per true millimetre would be $-0.01049/0.9807 = -0.01069$ mm^{-1} instead of -0.01049 mm^{-1}. This change (-0.00021 mm^{-1}) is far smaller than the uncertainty of the slope (± 0.00851 mm^{-1}), so the visibility-analysis conclusions do not depend on this possible calibration error. The backlash estimate from the He–Ne run (an intercept of -1.33 fringes, i.e. about 0.42 μm) is likewise negligible compared with the 0.2 mm steps in the Part 2 series. The reading uncertainty of z in that series (0.007 mm, Section 6.3.6) corresponds to about 26 fringes of the Part 2 source, showing that the coarse-knob reading fixes z only on a scale much coarser than a fringe.

7.4 Fringe contrast during wavelength measurement

The wavelength measurement required fringes that could be counted reliably (about 31 per division, regular to within about ± 1), which needs reasonable contrast. Contrast was not measured during wavelength determination, so no quantitative link to the visibilities measured in the second experiment can be made. The measured visibility values (V_{avg} from 0.105 to 0.684) show that recorded contrast can vary widely between photographs, which is a reminder that counting is easiest when the fringes are sharp.

7.5 Alignment, source and method

The wavelength measurement used circular fringes (parallel film), counted at the centre. Many photographs from the Part 2 visibility measurement show elliptical or curved fringes, indicating a slightly wedge-shaped film or an oblique view; the alignment quality was therefore not identical in the two experiments. The He–Ne wavelength result is not used to calculate the Part 2 visibility values; the stated Part 2 wavelength is 533.3 nm. The fringe-counting method could also be applied to this source, but was not used here. Both experiments share practical limitations including backlash, knob-reading uncertainty, vibrations, and a single measurement run.

8 Limitations and uncertainty sources

8.1 Wavelength measurement

Random errors

- While counting, it was sometimes difficult to know whether the fringe had exactly reached the centre, so there could be a difference of about ±1 fringe in the count.
- Even small vibrations in the table or the instruments made the fringes move a little, which made it difficult to count them properly.
- Sometimes the fringes were bright, and sometimes they became very faint. Due to this and slight air movement, a few fringes were not clearly visible during the experiment.
- We were missing the fractional arrival/departure of fringe due to fixed rotational steps of knob.

Systematic errors

- Backlash in the knob when it starts to turn (suggested by the negative intercept).
- Systematic under-counting of quickly moving fringes.
- Calibration of the coarse scale (0.010 mm per division assumed exactly).
- Slight misalignment of M_1 and M_2', which makes the pattern off-centre and the centre harder to define.

8.2 Fringe-visibility analysis

1. **Single profile per photograph.** Only one line through the pattern was analysed; no window-based averaging over the whole pattern was done.
2. **Three peak–valley pairs per photograph.** The mean of only three values has a large uncertainty (δ_{scat}).
3. **Gamma encoding.** The luma is not linear in intensity; absolute visibilities are underestimated (Section 6.3.11).
4. **Unknown exposure and camera settings.** Not recorded for each photograph.
5. **Speckle and noise.** Present in all photographs; handled by smoothing, with residual effects in δ_{prop} and δ_{meth}.
6. **Unresolved fringes in four photographs** (flagged).
7. **Backlash and calibration of the mirror drive.** The coarse scale was assumed to be exactly 0.01 mm per division.
8. **No spectral-width measurement.** The source linewidth was not supplied, and the visibility data are too scattered to determine a reliable coherence envelope or intrinsic coherence length.
9. **Unequal step size.** The step in z changes from 0.2 to 0.4 mm at $z = 6.4$ mm, so the later part of the range is sampled more coarsely.
10. **Absolute path difference unknown.** The position of equal arm lengths was not recorded, so $2z$ is only the *change* of path difference.
11. **Small sample.** With $n = 42$ correlated points the statistical power to detect a slope of the size observed is low.

9 Recommendations for a repeat experiment

9.1 Wavelength measurement

- Turn the fine knob through one complete revolution before counting starts; this removes most of the backlash error (do not touch the coarse knob).
- Turn the knob slowly so that you do not miss any fringe.
- Always turn the knob in the same direction during the entire measurement to avoid backlash error.
- Do not look directly into the laser beam, and do not look at the fringes by reflecting the beam into the eye with a mirror.
- Do not touch any optical surface (lens, mirror) with bare or oily hands.
- Keep the table free from vibrations and tighten all mounts so that the fringes remain stable and steady.
- Count at a fixed spot (centre is preferred), and ideally have a second observer count independently.

9.2 Fringe-visibility analysis

- Turn the knob slowly in one direction only and record the coarse and fine readings for every photograph.
- Fix the camera position, zoom, focus, exposure and screen distance for the whole series; use manual exposure and avoid saturation. Save images without compression and, if possible, as raw files so that the intensity is linear.
- Keep the screen illuminated only by the interference pattern (dark room) and avoid vibrations.
- Analyse the same fixed region of the pattern in every photograph, and use windows over the pattern to obtain many local visibilities per photograph.
- Record the alignment (tilt) for every photograph, or realign to circular fringes before each photograph.
- Take several photographs at each position to estimate the repeatability of V directly.
- Measure visibility at finer, equal steps in z and extend the range, while recording the source linewidth or enough calibrated visibility points to resolve the coherence envelope.
- Avoid looking directly at the bright source and do not touch optical surfaces.

10 Conclusion

Wavelength measurement

The Michelson interferometer was aligned with a He–Ne laser to obtain concentric circular fringes. For a mirror travel of 0.100 mm (from 0.173 mm to 0.273 mm), a total of 310 fringes were counted, giving

$$\lambda = 645.2 \pm 3.1 \text{ nm},$$

which is 12.4 nm (1.95%) above the given value of wavelength in the lab manual, 632.8 nm. A linear fit of the cumulative fringe count gave 643.2 ± 0.7 nm (slope 3109.7 ± 3.5 mm^{-1}, $R^2 = 0.99999$), in agreement with the direct value. Leaving out the first step changes the result only to 642.9 ± 2.3 nm. The measured wavelength is thus slightly but consistently high, which points to a small systematic effect such as fringe under-counting, backlash or scale calibration. The present data cannot determine which one. Within about 2%, the experiment verifies that $\lambda = 2\Delta d/N$ and demonstrates the use of the Michelson interferometer as a precise wavelength-measuring instrument.

Fringe-visibility analysis

Visibility was measured from 42 photographs of Michelson fringes taken at mirror displacements $z = 0$ to 10.0 mm. The average visibility per photograph, from three peak–valley pairs on a luma

profile, has mean 0.350 and ranges from 0.105 to 0.684; each value is known to about ±0.036 from noise, pair-to-pair scatter and analysis settings. A linear fit gives

$$V_{\text{avg}} = (0.395 \pm 0.044) + (-0.01049 \pm 0.00851)\, z \qquad (z \text{ in mm}),$$

with $r = -0.19$, $R^2 = 0.037$ and $p = 0.23$. The slope is negative, corresponding to a fall of about 0.105 over 10 mm, but its 95% confidence interval ($[-0.0277, 0.0067]$ mm^{-1}) includes zero, and weighted, non-parametric, bootstrap, permutation and leave-one-out analyses all agree that it is not statistically significant. A straight line describes the data poorly: the visibility oscillates with z in waves much larger than the measurement uncertainty ($\chi^2_\nu = 29.8$), and the residuals are correlated. Converting the luma to linear intensity changes the absolute level of the visibility but not the trend. The source linewidth was not measured, and the data do not resolve a reproducible coherence envelope. At $z = 10.0$ mm, Photo 42 still has luma visibility 0.374; if the scan began near equal arm lengths, this indicates an operational coherence-length scale of at least about 20 mm. Since the absolute zero-path-difference position and source linewidth were not measured, this is a conditional lower-bound estimate, not a precise coherence length. Since the exposure, alignment and other conditions were not recorded together with z, the data cannot identify the cause of the variations of the visibility, and the experiment gives only a weak indication of a decrease of contrast with increasing mirror displacement.

Overall

With the same interferometer, wavelength measurement gave regular, countable fringes and a wavelength within about 2% of the accepted value, while fringe-visibility analysis gave photographs whose fringe visibility was moderate on average and strongly variable, with only a weak, non-significant downward drift with mirror displacement. The results are consistent, with neither contradicting the other, and they remain independent measurements (Section 7): the possible calibration error suggested by wavelength measurement ($\approx 2\%$) changes the fringe-visibility analysis slope by far less than its uncertainty, and the fringe scale established by the laser measurement shows that the steps used for visibility analysis are too coarse to resolve individual fringes between photographs.

References

[1] Michelson's Interferometer, laboratory manual, last updated September 2023, © NISER.

[2] Supplier manual (listed as a reference in the laboratory manual).

[3] F. A. Jenkins and H. E. White, *Fundamentals of Optics*, 4th ed., McGraw–Hill, 1976.